

Comb-bounded crowns and bi-Esakia representability of well-ordered rooted trees

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Abstract

We give a sufficient condition for an abstract well-ordered rooted tree to admit a bi-Esakia topology. Let X have root r , and for every cover a of r write $Q_a = \uparrow a$ for the corresponding first-level crown. We prove that X is bi-Esakia representable whenever all Q_a are finite and their order duals uniformly omit one fixed finite co-tree comb.

The proof separates topology from finite order theory. First, we prove a synchronization theorem for arbitrarily many finite posets amalgamated over a common finite downset. If the petals fall into finitely many classes and every petal admits a stem-exact surjective bi-p-morphism onto an actual anchor petal, then the amalgam carries a bi-Esakia topology. The topology is a finite sum of one-point compactifications of anchor fibers and also arises as the inverse-limit topology of the finite-petal quotients. Second, a better-partial-order theorem of Martins supplies finitely many actual minimal anchor types for every uniformly comb-bounded family of finite crowns. The resulting class allows arbitrary root width, no uniform bound on crown size, and no uniform depth bound. We also explain why a known infinite antichain of finite co-trees obstructs the finite-anchor method without providing a nonrepresentability theorem.

1 Introduction

Esakia duality represents Heyting algebras by ordered Stone spaces in which the down-closure of every clopen set is clopen. Bi-Heyting algebras require the same topology to support both directions: both down-closures and up-closures of clopen sets must remain clopen. Consequently, showing separately that a poset and its order dual are Esakia representable does not by itself produce a bi-Esakia topology on the original poset.

Fornasiero and Moraschini characterized the Esakia-representable well-ordered forests satisfying the appropriate chain-supremum condition [4]. Their recursive topology is intrinsically one-sided, and the corresponding bi-Esakia representability problem remains subtler. Tasiou proved inverse-limit closure for bi-Esakia spaces in her master's thesis [1, Theorem 4.2.6], synchronized arbitrary disjoint copies of a fixed finite connected poset [1, Theorem 4.3.4][2, Theorem 4.6], and represented the finite stages of a countable inverse system as

the zigzag image-finite part of a larger bi-Esakia space [1, Theorem 4.3.8][2, Theorem 4.3]. The present question is different: the finite rooted pieces need not be isomorphic, and they meet at a common root, or more generally at a common finite stem.

Our first result isolates the required topological construction. Suppose finite petals P_i share a finite downset K . If they can be partitioned into finitely many groups so that every petal in a group maps by a surjective bi-p-morphism onto an actual anchor petal, with the map fixing K and taking no external point into K , then the amalgam is bi-Esakia representable. The topology synchronizes all fibers lying over the same anchor point. This extends the point-synchronization idea for isomorphic copies to finitely many nonisomorphic quotient types and a common stem.

The second result applies this construction to trees. Martins proved that finite co-trees omitting a fixed finite comb form a better partial order under the surjective bi-p-image relation [5, Proposition 4.10]. Thus, among the finite crowns that actually occur above a root, there are only finitely many minimal quotient types. Choosing actual crowns as representatives gives exactly the anchors required by the synchronization theorem.

Theorem A. Let X be a well-ordered rooted tree with root r . For each $a \in \text{Cov}_X(r)$, put $Q_a = \uparrow_X a$. Assume that

1. every Q_a is finite; and
2. for some fixed $n \geq 1$, the finite tree-comb $H_n = C_n^{\text{op}}$ does not order-embed into any Q_a , where C_n is the finite co-tree n -comb of [5].

Then the underlying poset X admits a bi-Esakia topology.

The well-ordered hypothesis is essential to the statement as written. Indeed, let

$$X = \{0\} \cup (\mathbb{Q} \cap (0, 1))$$

with the usual order. It is a rooted tree whose root has no cover, so the crown assumptions are vacuous. It has no gaps and is not even Priestley representable by [4, Proposition 2.5]. The first frozen version of Theorem A omitted the well-ordered hypothesis and was therefore false. Section 4 uses well-ordering to prove that the first-level crowns exhaust all nonroot points.

The conclusion covers trees of arbitrary root width. The crown sizes need not have a common bound, and their depths may be unbounded. The uniform restriction is order-theoretic: the family omits one fixed comb.

2 Preliminaries

All posets are ordered by $\leq$. For x in a poset P , write $\uparrow_P x$ and $\downarrow_P x$ for its principal up- and downsets. A tree is a poset with a least element in which every principal downset is a chain. It is well-ordered if every principal downset is a well-ordered chain.

A map $f : P \rightarrow Q$ is a *surjective bi-p-morphism* if it is surjective and order preserving and, for every $x \in P$,

$$f[\uparrow_P x] = \uparrow_Q f(x), \quad f[\downarrow_P x] = \downarrow_Q f(x). \quad (2.1)$$

Equivalently, it satisfies both the bounded and co-bounded back conditions. These conditions imply, for every $D \subseteq Q$,

$$\downarrow_P f^{-1}(D) = f^{-1}(\downarrow_Q D), \quad \uparrow_P f^{-1}(D) = f^{-1}(\uparrow_Q D). \quad (2.2)$$

A bi-Esakia space is an ordered Stone space whose clopen sets have clopen down- and up-closures. We use the equivalent criterion from [3, Theorem 2.12]: besides the clopen-cone conditions, it is enough in the present Stone setting to verify that every principal upset is closed.

For each positive integer n , let C_n be the finite co-tree with points

$$\{c_1, \dots, c_n, c'_1, \dots, c'_n\},$$

whose order is generated by

$$c_1 < c_2 < \dots < c_n, \quad c'_i < c_i \quad (1 \leq i \leq n). \quad (2.3)$$

This is the n -comb of [5, Figure 6], and we put $H_n = C_n^{\text{op}}$. We define the height of a finite poset to be the maximum number of points in a chain. Thus $C_1 = H_1^{\text{op}}$ is a two-point chain, H_n has height $n + 1$, and H_2 is branched and contains a three-point chain.

For finite co-trees, write $M \preceq_p Q$ when there exists a surjective bi-p-morphism $Q \rightarrow M$. Martins [5] proves that, for each fixed n , the class

$$\mathcal{T}_n = \{Y : Y \text{ is a finite co-tree and } C_n \text{ does not order-embed into } Y\} \quad (2.4)$$

is a better partial order under $\preceq_p$. In particular, it is a well partial order.

3 Synchronization over a common finite stem

We first prove the topological part independently of trees.

Theorem 3.1 (finite-anchor synchronization). *Let K be a finite poset and let G be finite. For each $g \in G$, let I_g be an arbitrary index set with a distinguished anchor $0_g \in I_g$. Suppose that finite posets $P_{g,i}$ satisfy the following conditions and that their amalgam is nonempty.*

1. *Every $P_{g,i}$ contains the same copy of K , and K is a downset of $P_{g,i}$.*
2. *Distinct petals meet exactly in K , and external points belonging to distinct petals are incomparable.*
3. *For every $i \in I_g$ there is a surjective bi-p-morphism*

$$f_{g,i} : P_{g,i} \longrightarrow P_{g,0_g}$$

which is the identity on K and satisfies $f_{g,i}^{-1}(K) = K$. For the anchor, take the identity map.

Let X be the amalgam of all petals over K . Then X admits a bi-Esakia topology.

Proof. For $g \in G$ and $a \in P_{g,0_g} \setminus K$, define the synchronized fiber

$$E_{g,a} = \{a\} \cup \bigcup_{i \in I_g \setminus \{0_g\}} f_{g,i}^{-1}(a). \quad (3.1)$$

Give $E_{g,a}$ the one-point compactification topology in which every point other than a is isolated and a is the unique possible nonisolated point. Give K the discrete topology. Since G , K , and every anchor petal are finite, X is the finite topological sum

$$X = K \dot{\cup} \bigcup_{g \in G} \bigcup_{a \in P_{g,0_g} \setminus K} E_{g,a}, \quad (3.2)$$

and is therefore a compact Stone space.

A set $U \subseteq X$ is clopen precisely when, for each synchronized fiber, its trace is finite if the anchor is absent and cofinite if the anchor is present. Because the number of anchors is finite and every individual fiber $f_{g,i}^{-1}(a)$ is finite, there is for each g a finite exceptional set $J_g \subseteq I_g$ such that

$$U \cap P_{g,i} = f_{g,i}^{-1}(U \cap P_{g,0_g}) \quad (i \notin J_g). \quad (3.3)$$

Conversely, this eventual inverse-image condition characterizes clopen subsets.

We show that the down-closure of U has the same form. Fix a nonexceptional petal and abbreviate

$$U_i = U \cap P_{g,i}, \quad U_0 = U \cap P_{g,0_g}, \quad T = (\downarrow_X U) \cap K.$$

No external point is comparable across distinct petals, K is a common downset, and $f_{g,i}^{-1}(T) = T$. Hence Equation (2.2) gives

$$\begin{aligned} (\downarrow_X U) \cap P_{g,i} &= \downarrow_{P_{g,i}} U_i \cup T \\ &= f_{g,i}^{-1}(\downarrow_{P_{g,0_g}} U_0 \cup T) \\ &= f_{g,i}^{-1}((\downarrow_X U) \cap P_{g,0_g}). \end{aligned} \quad (3.4)$$

The up-closure cannot move from an external point into another petal, while the common influence of $U \cap K$ is already present in every inverse image. Thus

$$\begin{aligned} (\uparrow_X U) \cap P_{g,i} &= \uparrow_{P_{g,i}} U_i \\ &= f_{g,i}^{-1}(\uparrow_{P_{g,0_g}} U_0) \\ &= f_{g,i}^{-1}((\uparrow_X U) \cap P_{g,0_g}). \end{aligned} \quad (3.5)$$

Only the finitely many exceptional finite petals and the finite set K can deviate from (3.4) or (3.5). Therefore $\downarrow_X U$ and $\uparrow_X U$ are clopen.

It remains to check principal upsets. If $x \in K$, stem-exactness and the co-bounded back condition imply that each synchronized fiber lies either wholly inside $\uparrow_X x$ or wholly outside it. Hence $\uparrow_X x$ is clopen. If $x \notin K$, then $\uparrow_X x$ lies inside one finite petal and is finite; it is closed because the Stone space is T_1 . The criterion from [3, Theorem 2.12] now shows that X is bi-Esakia. $\square$

Proposition 3.2 (inverse-limit realization). *The topology in Theorem 3.1 is the inverse-limit topology of the finite amalgams obtained by retaining finitely many petals and collapsing every unretained petal onto its anchor.*

Proof. Let

$$\mathcal{D} = \prod_{g \in G} \{F_g \in [I_g]^{<\omega} : 0_g \in F_g\}, \quad (3.6)$$

ordered by coordinatewise inclusion. It is directed because G is finite and coordinatewise finite unions give common upper bounds. For $F \in \mathcal{D}$, let X_F retain exactly the petals indexed by F_g in each group, and give this finite poset its discrete bi-Esakia topology. If $F \leq E$, define

$$\pi_{EF} : X_E \longrightarrow X_F \quad (3.7)$$

to be the identity on K and on petals already retained by F , and to use $f_{g,i}$ on every petal indexed by $E_g \setminus F_g$. The three possible locations of a point—stem, retained petal, or collapsed petal—show directly that both back conditions hold. The definitions also give

$$\pi_{FF} = \text{id}_{X_F}, \quad \pi_{LF} = \pi_{EF} \circ \pi_{LE} \quad (F \leq E \leq L). \quad (3.8)$$

Thus $(X_F, \pi_{EF})_{F \in \mathcal{D}}$ is an inverse system of finite posets and surjective bi-p-morphisms.

Define the canonical quotient $q_F : X \rightarrow X_F$ by retaining every F_g -petal and applying $f_{g,i}$ to every other petal. Then $q_F = \pi_{EF} \circ q_E$ whenever $F \leq E$, so

$$\eta : X \longrightarrow \varprojlim_{F \in \mathcal{D}} X_F, \quad \eta(x) = (q_F(x))_F, \quad (3.9)$$

is well defined.

Let $(x_F)_F$ be compatible, and inspect the least stage containing only the anchors. If its coordinate belongs to K , stem-exactness forces every higher coordinate, and hence every coordinate, to be the same stem point. Otherwise that coordinate is an external anchor point. Either every coordinate remains that anchor, or at some stage F a retained nonanchor petal contains an external point y . Above F , compatibility and identity on retained petals lock every coordinate to y . For an incomparable stage H , a common upper bound $L \geq F, H$ uniquely determines the H -coordinate by projection. Two distinct petals cannot both be selected, since a stage retaining both would have to project to two different external points. Thus η is surjective. It is also injective. For $x \neq y$, choose a finite stage retaining the nonanchor petals to which the external points among x, y belong; there are at most two such petals. Identity on retained petals separates two external points, while stem-exactness separates a point of K from every distinct point. Hence $q_F(x) \neq q_F(y)$.

For topology, $q_F^{-1}(\{k\}) = \{k\}$ for $k \in K$, by stem-exactness. The preimage of a nonanchor retained point is an isolated singleton. The preimage of an anchor point a contains a and the full a -fiber in every unretained petal, while it differs only within finitely many retained finite fibers. Thus every q_F is continuous for the synchronized topology, so that topology is at least as fine as the cylinder topology.

Conversely, let $E_{g,a} \setminus R$ be a basic synchronized neighborhood of an anchor, where R is finite and does not contain a . Choose F retaining every petal that contains a point of R , and put

$$V = \{a\} \cup ((E_{g,a} \cap X_F) \setminus R). \quad (3.10)$$

Since X_F is discrete, V is open, and direct inspection gives

$$q_F^{-1}(V) = E_{g,a} \setminus R. \quad (3.11)$$

The isolated stem and nonanchor points were already realized as singleton cylinders. These sets form a base for the finite topological sum (3.2), so the synchronized topology is also no finer than the cylinder topology. Hence the two topologies agree. Since X is compact and the inverse limit of finite discrete spaces is Hausdorff, the continuous bijection η is a homeomorphism. $\square$

4 Comb-bounded crowns

We now prove Theorem A.

Lemma 4.1 (root-crown decomposition). *Let X be a well-ordered rooted tree with root r . Then*

$$X = \{r\} \dot{\cup} \bigcup_{a \in \text{Cov}_X(r)} \uparrow_X a, \quad (4.1)$$

and external points from distinct crowns are incomparable.

Proof. For $x > r$, the nonempty interval $(r, x]$ is a subset of the well-ordered chain $\downarrow x$ and therefore has a least element a . No point lies strictly between r and a , so a covers r , and $x \in \uparrow a$. If two different covers a, b both lie below x , the chain $\downarrow x$ makes them comparable, contradicting that both cover r . The same argument rules out comparability between external points in different crowns. $\square$

Theorem 4.2 (comb-bounded crown theorem). *Under the hypotheses of Theorem A, X is bi-Esakia representable.*

Proof. Let $A = \text{Cov}_X(r)$ and $Q_a = \uparrow_X a$. By Lemma 4.1, the sets $P_a = \{r\} \cup Q_a$ are finite petals amalgamated over the common downset $K = \{r\}$.

Every dual crown Q_a^{op} belongs to $\mathcal{T}_n$. Let S be the set of isomorphism types that actually occur among these dual crowns. For each $s \in S$, well-foundedness gives a minimal element of the nonempty set $S \cap \downarrow s$. It is globally minimal in S , since any smaller element of S would also lie in $S \cap \downarrow s$. The global minimal elements of S form an antichain and are therefore finite. Choose, from the original tree, one actual crown index a_j for each minimal isomorphism type and put

$$M_j = Q_{a_j}^{\text{op}} \quad (j = 1, \dots, t). \quad (4.2)$$

Hence, for every $a \in A$, there is a $j = j(a)$ and a surjective bi-p-morphism

$$g_a : Q_a^{\text{op}} \longrightarrow M_j. \quad (4.3)$$

After transporting along an isomorphism if necessary, the target is the chosen actual representative. Taking order duals preserves both back conditions, so

$$g_a^{\text{op}} : Q_a \longrightarrow Q_{a_j} \quad (4.4)$$

is again a surjective bi-p-morphism.

Extend (4.4) to $f_a : P_a \rightarrow P_{a_j}$ by setting $f_a(r) = r$. This map is root-exact. At the root, the up-back condition follows from surjectivity of (4.4), while the only lower point is the root itself. Inside a crown, both back conditions come from (4.4), except that a lower target equal to the common root is lifted by the root. Therefore f_a is a surjective bi-p-morphism and $f_a^{-1}(\{r\}) = \{r\}$.

There are only finitely many anchors P_{a_j} . Theorem 3.1 applies and supplies a bi-Esakia topology on X . $\square$

5 Consequences

Corollary 5.1 (three-level trees). *Every well-ordered rooted tree of height at most three in which each first-level point has finitely many immediate successors is bi-Esakia representable, with no restriction on the cardinality of the first level.*

Proof. Each crown is a finite star and therefore has height at most two. Since H_2 has height three, no crown contains H_2 . Apply Theorem 4.2. $\square$

Corollary 5.2 (uniform finite height). *Let X be a well-ordered rooted tree whose first-level crowns are finite. If the heights, measured by the maximum number of points in a chain, have a common finite bound h , then X is bi-Esakia representable. Crown cardinalities need not be uniformly bounded.*

Proof. Choose n with $n + 1 > h$. Since H_n has height $n + 1$, it cannot embed into any crown. Apply Theorem 4.2. $\square$

Corollary 5.3 (unbounded depth). *Theorem 4.2 applies to families of crowns with unbounded depth whenever they uniformly omit one fixed comb. In particular, arbitrary families of finite chains are covered.*

Proof. The fixed poset H_2 is branched, so it does not embed into a chain, independently of the chain length. $\square$

6 The finite-anchor boundary

The uniform comb bound is not a decorative assumption in the proof. Bezhanishvili, Martins, and Moraschini record a countably infinite antichain of finite co-trees under the surjective bi-p-image relation [3, Proposition 4.15]. Taking order duals gives an antichain of finite rooted trees.

If all these rooted trees are placed as crowns above a common root, the resulting family has no finite set of minimal anchor types generated by the WPO argument. Thus Theorem 3.1 cannot be applied by the present finite-group reduction.

This is only a limitation of the method. It does not prove that the common-root tree built from the antichain lacks every possible bi-Esakia topology. A genuine negative result would require a topology-independent obstruction, or a proof that any bi-Esakia topology forces a finite quotient comparability absent from the antichain. Neither statement is established here.

7 Relation to prior work

Tasiou’s inverse-limit theorem [1, Theorem 4.2.6] supplies the general categorical background. Her synchronization result for arbitrary copies of one finite connected poset [1, Theorem 4.3.4][2, Theorem 4.6] is the closest explicit construction; [1, Theorem 4.3.8][2, Theorem 4.3] instead realizes the finite stages of a countable inverse system as the zigzag image-finite part of a larger space. Theorem 3.1 differs in three coupled features: the pieces may be nonisomorphic, only finitely many actual quotient anchors are required, and all pieces may share a finite downset. The inverse-limit mechanism itself is not new; the contribution is the stem-exact finite-anchor criterion and its application to a natural infinite class of bare trees.

Fornasiere and Moraschini [4] address Esakia representability of well-ordered forests. Their result motivates the object class but does not provide a common bi-Esakia topology. Martins [5] proves the BPO theorem used in Section 4 but does not discuss bi-Esakia topologies on the common-root amalgams. The antichain in [3] explains why the same finite-anchor argument cannot simply discard the comb restriction.

The present result is therefore a sufficient theorem, not a classification. It neither proves that the comb condition is necessary nor settles bi-Esakia representability for arbitrary well-ordered trees or forests.

8 Open directions

The most direct continuation is to replace the finite anchor family in Theorem 3.1 by a compact parameter space of anchor types. Such a construction would have to control clopen up- and down-closures across infinitely many groups and prevent new inverse-limit points. The known antichain shows that order-theoretic minimality alone cannot provide this compactness.

A different route is negative: identify a topology-independent invariant that prevents a common-root amalgam from being bi-Esakia representable. At present, the antichain supplies no such invariant.

Other extensions—an infinite common stem, infinite petals, or arbitrary forest components—each break a separate finiteness step in Theorem 3.1 and should be treated as new problems rather than formal corollaries.

9 AI-assisted research and writing disclosure

This work was developed with AI assistance for literature search, proof exploration, adversarial checking, computation, and drafting. Carptopus is the responsible author and independently controls the mathematical claims and public version. OpenAI Codex is a research and writing tool, not an author.

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